Yarim va chorak doira yuzi

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Yarim doiraning diametri 3 m. Uning yuzini toping (π = 3,14).**

A) 7,065 m² **B)** 3,5325 m² **C)** 14,13 m² **D)** 7,71 m²

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2. **Devor soatining siferblati doira shaklida, diametri 9,8 dm. Uning 12 dan 3 gacha bo'lgan qismi sariqqa bo'yalgan. Bo'yalgan qismning yuzini toping (π = $\frac{22}{7}$).**

A) 75,46 dm² **B)** 18,865 dm²

C) 7,7 dm² **D)** 37,73 dm²

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3. **Dumaloq pishloqning radiusi 4,2 cm. Undan chorak bo'lak kesib olindi. Qolgan pishloqning ustki yuzi qancha (π = $\frac{22}{7}$)?**

A) 41,58 cm² **B)** 13,86 cm² **C)** 27,72 cm² **D)** 55,44 cm²

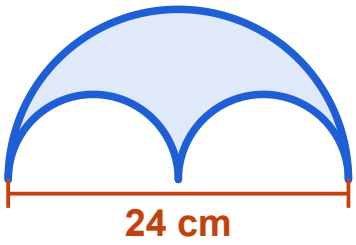
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4. **Bog'dagi yarim doira shaklidagi sahnaning radiusi 6,5 m. Sahna yuzini kalkulyatordagi π tugmasi bilan hisoblang va yuzdan birlar xonasigacha yaxlitlang.**

A) 66,33 m² **B)** 66,36 m² **C)** 66,37 m² **D)** 132,73 m²

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5. **Ko'prik maketi yarim doira shaklidagi kartondan yasaldi: uning diametri 24 cm, pastidan ikkita bir xil yarim doira shaklidagi ark teshigi o'yildi (rasm). Qolgan kartonning yuzi qancha (π = 3,14)?**

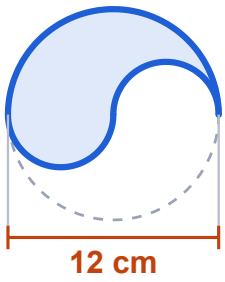


A) 226,08 cm² **B)** 339,12 cm²

C) 169,56 cm² **D)** 113,04 cm²

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6. **Diametri 12 cm bo'lgan dumaloq nishon ikki rangga bo'yalgan. Ranglar chegarasi — ikkita bir xil yarim aylana (rasm). Bo'yalgan qismning yuzi qancha (π = 3,14)?**



A) 84,78 cm² **B)** 28,26 cm²

C) 56,52 cm² **D)** 113,04 cm²

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7. **Yarim doira shaklidagi gilamning diametri 2 m. 1 m² gilam 80 000 so'm turadi. Gilam necha so'm turadi (π = 3,14)?**

A) 125 600 so'm **B)** 62 800 so'm

C) 251 200 so'm **D)** 502 400 so'm

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8. **Uchta bir xil chorak doira va bitta yarim doira bor, hammasining radiusi 2 dm. Ularning yuzlari yig'indisi qancha (π = 3,14)?**

A) 12,56 dm² **B)** 18,84 dm² **C)** 6,28 dm² **D)** 15,7 dm²

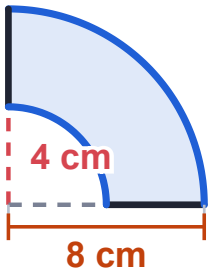
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9. **Radiusi 6 cm bo'lgan yarim doiraning yuzi xuddi shunday radiusli chorak doiraning yuzidan qancha katta (π = 3,14)?**

A) 56,52 cm² **B)** 84,78 cm² **C)** 9,42 cm² **D)** 28,26 cm²

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10. **Radiusi 8 cm bo'lgan chorak doiradan markazi bir xil, radiusi 4 cm bo'lgan chorak doira kesib olindi. Bo'yalgan qismning yuzini toping (π = 3,14).**



A) 62,8 cm² **B)** 12,56 cm²

C) 37,68 cm² **D)** 50,24 cm²

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11. **Bog'bon diametri 10 m bo'lgan yarim doira shaklidagi gulzorning har kvadrat metriga 4 tadan ko'chat o'tqazdi. U jami nechta ko'chat o'tqazdi (π = 3,14)?**

A) 40 ta **B)** 628 ta **C)** 157 ta **D)** 314 ta

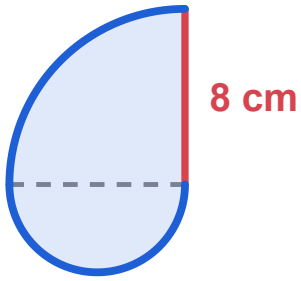
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12. **Diametri 4 cm bo'lgan ikkita yarim doira va radiusi 2 cm bo'lgan to'rtta chorak doira bor. Ularning yuzlari yig'indisi qancha (π = 3,14)?**

A) 50,24 cm² **B)** 12,56 cm² **C)** 25,12 cm² **D)** 37,68 cm²

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13. **Shakl chorak doira va yarim doiradan tuzilgan (rasm). Uning yuzini toping (π = 3,14).**



A) 62,8 cm² **B)** 150,72 cm²

C) 37,68 cm² **D)** 75,36 cm²

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14. **Sevara diametri 10 cm bo'lgan yarim doiraning yuzini shunday topdi: $\frac{1}{2}$ · 3,14 · 10 · 10 = 157 cm². To'g'ri javob qancha?**

A) 19,625 cm² **B)** 78,5 cm²

C) 39,25 cm² **D)** 157 cm²

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15. **Yarim doiraning yuzi 9,42 m². Radiusi xuddi shunday bo'lgan chorak doiraning yuzi qancha?**

A) 18,84 m² **B)** 9,42 m² **C)** 2,355 m² **D)** 4,71 m²

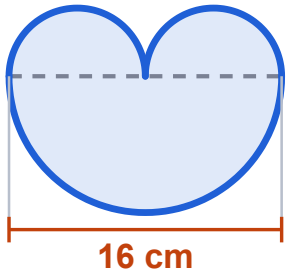
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16. **Doira shaklidagi gulzorning diametri 8 m. Uning choragiga atirgul, yarmiga lola, qolgan qismiga rayhon ekildi. Rayhon ekilgan joyning yuzi qancha (π = 3,14)?**

A) 37,68 m² **B)** 25,12 m² **C)** 50,24 m² **D)** 12,56 m²

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17. **Bulut rasmi: diametri 16 cm bo'lgan yarim doira ustiga diametri 8 cm bo'lgan ikkita yarim doira qo'yilgan. Shaklning yuzini toping (π = 3,14).**



A) 150,72 cm²
B) 200,96 cm²
C) 100,48 cm²
D) 50,24 cm²

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18. **Yarim doira shaklidagi stol ustining diametri 1,2 m. Uni qoplash uchun 1 m² laminat 50 000 so'm. Necha so'mlik laminat kerak (π = 3,14)?**

A) 113 040 so'm
B) 56 520 so'm
C) 28 260 so'm
D) 14 130 so'm

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19. **Yarim doira shaklidagi deraza o'rtadan tik ramka bilan ikkita bir xil oynaga bo'lingan. Derazaning diametri 1,4 m. Har bir oynaning yuzi qancha (π = $\frac{22}{7}$)?**

A) 3,08 m²
B) 1,54 m²
C) 0,77 m²
D) 0,385 m²

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20. **Radiusi 10 cm bo'lgan chorak doiraning ikkala to'g'ri tomoniga tashqaridan diametri 10 cm bo'lgan yarim doiralar yopishtirildi. Hosil bo'lgan shaklning yuzini toping (π = 3,14).**

A) 157 cm²
B) 314 cm²
C) 117,75 cm²
D) 78,5 cm²

Javoblar

Yarim va chorak doira yuzi · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	3,5325 m²
2	**B**	18,865 dm²
3	**A**	41,58 cm²
4	**C**	66,37 m²
5	**D**	113,04 cm²
6	**C**	56,52 cm²
7	**A**	125 600 so'm
8	**D**	15,7 dm²
9	**D**	28,26 cm²
10	**C**	37,68 cm²

№	Javob	Qiymati
11	**C**	157 ta
12	**C**	25,12 cm²
13	**D**	75,36 cm²
14	**C**	39,25 cm²
15	**D**	4,71 m²
16	**D**	12,56 m²
17	**A**	150,72 cm²
18	**C**	28 260 so'm
19	**D**	0,385 m²
20	**A**	157 cm²

Qisqa yechimlar

1. Radius: 3 : 2 = 1,5 m. r² = 1,5 · 1,5 = 2,25.
$\frac{1}{2}$ · 3,14 · 2,25 = **3,5325 m²**.
(7,71 m — bu perimetr: yoy + diametr.)

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2. 12 dan 3 gacha — siferblatning choragi. Radius: 9,8 : 2 = 4,9 dm. Avval 4,9 : 7 = 0,7.
$\frac{1}{4}$ · 22 · 0,7 · 4,9 = **18,865 dm²**.
(75,46 — diametr radius deb olingan.)

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3. Qolgan qism — doiraning $\frac{3}{4}$ qismi.
Doira: $\frac{22}{7}$ · 4,2 · 4,2 = 22 · 0,6 · 4,2 = 55,44 cm².
$\frac{3}{4}$ qismi: 55,44 : 4 · 3 = **41,58 cm²**.

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4. π · 6,5 · 6,5 : 2 = 66,3661… m².
Uchinchi raqam 6, shuning uchun: **66,37 m²**.
(66,33 — π = 3,14 bilan chiqadi; 66,36 — yaxlitlanmay tashlangan.)

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5. Katta: r = 12 cm, $\frac{1}{2}$ · 3,14 · 144 = 226,08 cm².
Kichiklari: diametri 12 cm, r = 6 cm. Ikkalasi: 2 · $\frac{1}{2}$ · 3,14 · 36 = 113,04 cm².
226,08 − 113,04 = **113,04 cm²**.

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6. Chapdan qo'shilgan kichik yarim doira o'ngdan olingan yarim doiraga teng.
Bo'yalgan soha katta yarim doiraga teng: r = 6 cm, $\frac{1}{2}$ · 3,14 · 36 = **56,52 cm²**.

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7. Radius: 2 : 2 = 1 m. Yuzi: $\frac{1}{2}$ · 3,14 · 1 · 1 = 1,57 m².
Narxi: 1,57 · 80 000 = **125 600 so'm**.

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8. Ulushlar: $\frac{3}{4} + \frac{1}{2} = \frac{3}{4} + \frac{2}{4} = \frac{5}{4}$ doira.
Doira: 3,14 · 2 · 2 = 12,56 dm².
12,56 : 4 · 5 = **15,7 dm²**.

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9. Yarim doira: $\frac{1}{2}$ · 3,14 · 36 = 56,52 cm². Chorak: $\frac{1}{4}$ · 3,14 · 36 = 28,26 cm².
56,52 − 28,26 = **28,26 cm²**.

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10. Katta chorak: $\frac{1}{4}$ · 3,14 · 64 = 50,24 cm². Kichigi: $\frac{1}{4}$ · 3,14 · 16 = 12,56 cm².
50,24 − 12,56 = **37,68 cm²**.

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11. Radius: 10 : 2 = 5 m. Yuzi: $\frac{1}{2}$ · 3,14 · 25 = 39,25 m².
Ko'chat: 39,25 · 4 = **157 ta**.

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12. Hammasining radiusi 2 cm. 2 ta yarim — 1 ta doira, 4 ta chorak — yana 1 ta doira.
Doira: 3,14 · 2 · 2 = 12,56 cm². Ikkitasi: **25,12 cm²**.

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13. Chorak doira: r = 8 cm, $\frac{1}{4}$ · 3,14 · 64 = 50,24 cm².
Yarim doiraning diametri — chorakning radiusi, 8 cm. Uning radiusi 4 cm: $\frac{1}{2}$ · 3,14 · 16 = 25,12 cm².
50,24 + 25,12 = **75,36 cm²**.

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14. Sevara diametrni radius o'rniga qo'ydi. Radius: 10 : 2 = 5 cm.
$\frac{1}{2}$ · 3,14 · 5 · 5 = **39,25 cm²**.

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15. Chorak doira — yarim doiraning yarmi.
9,42 : 2 = **4,71 m²**.

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16. Qolgan qism: $1 - \frac{1}{4} - \frac{1}{2} = \frac{1}{4}$ doira.
Radius 4 m: $\frac{1}{4}$ · 3,14 · 16 = **12,56 m²**.

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17. Katta: r = 8 cm, $\frac{1}{2}$ · 3,14 · 64 = 100,48 cm².
Kichiklari: r = 4 cm, $2 \cdot \frac{1}{2}$ · 3,14 · 16 = 50,24 cm².
Ular qo'shiladi: 100,48 + 50,24 = **150,72 cm²**.

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18. Radius: 1,2 : 2 = 0,6 m. Yuzi: $\frac{1}{2}$ · 3,14 · 0,6 · 0,6 = 0,5652 m².
Narxi: 0,5652 · 50 000 = **28 260 so'm**.

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19. Har bir oyna — chorak doira, radiusi 1,4 : 2 = 0,7 m.
$\frac{1}{4} \cdot \frac{22}{7}$ · 0,7 · 0,7 = $\frac{1}{4}$ · 22 · 0,1 · 0,7 = **0,385 m²**.

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20. Chorak doira: $\frac{1}{4}$ · 3,14 · 100 = 78,5 cm².
Yarim doiralar: r = 5 cm, $2 \cdot \frac{1}{2}$ · 3,14 · 25 = 78,5 cm².
78,5 + 78,5 = **157 cm²**.
